

Thermodynamics of Hydrogen-Bonding Mixtures. 5. G^E , H^E , and TS^E and Zeotropy of Water + Acrylic Acid

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A stirred-flask ebulliometer was used to measure total vapor-pressure (PTx) data on nine mixtures of water + acrylic acid (and the pure components) between 45 and 112 °C and pressures from 10.00 to 101.325 kPa. These PTx data and the derived vapor–liquid equilibrium data show no azeotropic behavior. To verify that water + acrylic acid is a zeotrope at 101.325 kPa, dilute-solution ΔT data at 101.325 kPa were measured in a twin-arm Cottrell-lift-pump ebulliometer. Data on water + propanoic acid, with a known azeotrope at 101.325 kPa, were measured to test the experimental method. These dilute-solution measurements confirmed the known azeotrope for water + propanoic acid and conclusively demonstrated zeotropy for water + acrylic acid at 101.325 kPa. The system exhibits large positive deviations from ideality (derived $\gamma^\infty = 2.5$ –5.7) that decrease with increasing temperature. Equimolar G^E/T derived from fitted isothermal activity coefficient parameters decreases with increasing temperature, which predicts a positive H^E . The excess function data show that the system water + acrylic acid belongs to the class of mixtures where $G^E > 0$ and $H^E > 0$ (this includes other water + organic acid mixtures); TS^E shows a crossover from $TS^E > 0$ to $TS^E < 0$.

Introduction

Previous papers in this series^{1–4} have presented mixture excess thermodynamic data and vapor–liquid equilibria (VLE) for mixtures of diols,^{1,3} an alkane + alcohol,² and water + an alkylethylenamine.⁴ This paper presents mixture PTx VLE data and excess thermodynamic data for the system water + acrylic acid (2-propanoic acid, CAS No. 79-10-7). In addition, new azeotropic data for water + propanoic acid (CAS No. 79-09-4) were measured.

VLE data for water + acrylic acid are sparse: the Dortmund Databank (www.ddbst.de) gives five sets of VLE data;^{5–9} an azeotrope at 101.325 kPa and $y_1 = 0.9967$ is reported by Frolov et al.⁶ No single study has reported VLE data over a wide range of temperature and pressure.

Experimental Section

The water (HPLC grade) and propanoic acid (99.5+ mass %) were from Aldrich and were used as received; the acrylic acid was freshly prepared locally by The Dow Chemical Company Acrylate R&D department and had a purity of 99.5+ mass percent; except for one set of dilute-solution experiments, the acrylic acid was then inhibited with 0.1 mass % phenothiazine (CAS No. 92-84-2) to prevent polymerization. Mixtures were prepared gravimetrically with analytical balances.

Two ebulliometers were used:

(1) A 125 cm³ stirred-flask ebulliometer¹⁰ was used to measure PTx data for the pure components and nine mixtures from 0.1 to 0.9 mol fraction. The stirred flask was heated in a silicon oil bath that was thermostatted about 12 °C above the bubble point of the sample at each pressure. A condenser cooled to 5 °C provided connection to the manostat. Pressures were controlled with a Mensor model PCS 400 quartz manostat/manometer to ± 0.007 kPa. Temperatures on the ITS-90 scale were measured to ± 0.001 °C with a 100 Ω platinum resistance thermometer that had been calibrated by comparison with a Hart

Scientific SPRT. Bubble-point temperatures were measured at 10.00, 13.33, 26.66, 40.00, 53.33, 66.66, 79.99, and 101.325 kPa.

(2) Differences in the isobaric bubble-point temperature of pure water and dilute acid mixtures were measured in a twin-arm Cottrell vapor-lift pump ebulliometer,¹¹ which sprays slugs of equilibrated liquid and vapor upon a thermometer well at total reflux. Small mass increments of acid were injected into the ebulliometer using a gastight syringe; the temperature and pressure measurements were the same as in 1.

Results

Table 1 contains 88 PTx measurements on the pure components and nine mixtures; also shown are the derived vapor compositions and relative volatilities for the mixtures. The difference between x , the liquid-phase mole fraction, and z , the overall mole fraction charged to the ebulliometer, shows the magnitude of the vapor-holdup and condensed vapor-holdup correction computed as described previously.¹² The PT data for each mixture were fitted with a three-constant Antoine equation in order to interpolate isothermal Px data similar to the procedure described by Prengle and Palm.¹³ The Antoine constants and derived Pxy data for 75 °C are given in Table 2.

Figure 1 shows a comparison of the measured pure-component vapor-pressure data with literature data: the water data are compared to vapor pressure data from the NBS/NRC Steam Tables;¹⁴ the acrylic acid data are compared to vapor-pressure data found in the DIPPR databank.¹⁵ However, the mixture VLE data should be analyzed with the pure-component vapor-pressure data that were measured in the same apparatus and on the same chemicals used in the mixture study rather than using literature or handbook vapor pressure data.¹⁶

Values of G^E and y , the vapor-phase mole fraction, were computed by a Gauss–Newton nonlinear least-squares fit to the experimental mixture vapor pressures coupled with a bubble-point calculation during each iteration (Barker's method). The

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Table 1. *PTx* Data for Water (1) + Acrylic Acid (2)

<i>T</i> (°C)	<i>P</i> (kPa)	<i>z</i> ₁	<i>x</i> ₁	<i>y</i> ₁ (calc)	α (calc)	<i>T</i> (°C)	<i>P</i> (kPa)	<i>z</i> ₁	<i>x</i> ₁	<i>y</i> ₁ (calc)	α (calc)
79.97	10.00	0.0000	0.0000	0.0000		122.45	53.33	0.0000	0.0000	1.0000	
63.26	10.00	0.1003	0.0869	0.4909	10.13	105.97	53.33	0.1003	0.0910	0.4109	6.97
57.44	10.00	0.2005	0.1834	0.6619	8.71	98.34	53.33	0.2005	0.1875	0.5966	6.41
54.08	10.00	0.3008	0.2845	0.7425	7.25	93.87	53.33	0.3008	0.2875	0.6964	5.68
51.98	10.00	0.3999	0.3863	0.7904	5.99	90.70	53.33	0.3999	0.3883	0.7589	4.96
50.35	10.00	0.5000	0.4896	0.8252	4.92	88.44	53.33	0.5000	0.4908	0.8042	4.26
48.84	10.00	0.6007	0.5934	0.8543	4.02	86.48	53.33	0.6007	0.5941	0.8406	3.60
47.95	10.00	0.7131	0.7089	0.8845	3.14	85.15	53.33	0.7131	0.7094	0.8764	2.91
47.26	10.00	0.8103	0.8082	0.9132	2.50	84.18	53.33	0.8103	0.8084	0.9080	2.34
46.88	10.00	0.8998	0.8990	0.9476	2.03	83.59	53.33	0.8998	0.8991	0.9439	1.89
46.49	10.00	1.0000	1.0000	1.0000	1.00	83.33	53.33	1.0000	1.0000	0.9990	1.00
86.45	13.33	0.0000	0.0000	0.0000		129.13	66.66	0.0000	0.0000	1.0000	
69.67	13.33	0.1003	0.0876	0.4763	9.47	112.78	66.66	0.1003	0.0915	0.4011	6.65
63.61	13.33	0.2005	0.1841	0.6504	8.24	104.84	66.66	0.2005	0.1880	0.5882	6.17
60.09	13.33	0.3008	0.2850	0.7344	6.94	100.13	66.66	0.3008	0.2879	0.6905	5.52
57.85	13.33	0.3999	0.3866	0.7847	5.78	96.78	66.66	0.3999	0.3885	0.7550	4.85
56.15	13.33	0.5000	0.4898	0.8213	4.79	94.40	66.66	0.5000	0.4909	0.8018	4.19
54.58	13.33	0.6007	0.5935	0.8516	3.93	92.34	66.66	0.6007	0.5942	0.8392	3.57
53.62	13.33	0.7131	0.7090	0.8827	3.09	90.94	66.66	0.7131	0.7094	0.8757	2.89
52.90	13.33	0.8103	0.8082	0.9120	2.46	89.92	66.66	0.8103	0.8084	0.9077	2.33
52.50	13.33	0.8998	0.8990	0.9468	2.00	89.31	66.66	0.8998	0.8991	0.9436	1.88
52.14	13.33	1.0000	1.0000	1.0000	1.00	89.04	66.66	1.0000	1.0000	0.9990	1.00
103.38	26.66	0.0000	0.0000	0.0000		134.80	79.99	0.0000	0.0000	1.0000	
86.67	26.66	0.1003	0.0894	0.4427	8.09	118.57	79.99	0.1003	0.0919	0.3933	6.40
79.86	26.66	0.2005	0.1859	0.6232	7.25	110.44	79.99	0.2005	0.1884	0.5813	5.98
75.93	26.66	0.3008	0.2863	0.7151	6.26	105.47	79.99	0.3008	0.2882	0.6857	5.39
73.28	26.66	0.3999	0.3875	0.7714	5.34	101.94	79.99	0.3999	0.3886	0.7519	4.77
71.33	26.66	0.5000	0.4904	0.8123	4.50	99.46	79.99	0.5000	0.4910	0.7999	4.14
69.61	26.66	0.6007	0.5938	0.8456	3.75	97.31	79.99	0.6007	0.5942	0.8381	3.54
68.51	26.66	0.7131	0.7092	0.8791	2.98	95.86	79.99	0.7131	0.7094	0.8752	2.87
67.67	26.66	0.8103	0.8083	0.9097	2.39	94.77	79.99	0.8103	0.8084	0.9074	2.32
67.16	26.66	0.8998	0.8990	0.9451	1.93	94.14	79.99	0.8998	0.8991	0.9434	1.87
66.88	26.66	1.0000	1.0000	0.9990	1.00	93.87	79.99	1.0000	1.0000	0.9990	1.00
114.24	40.00	0.0000	0.0000	1.0000		142.47	101.325	0.0000	0.0000	1.0000	
97.66	40.00	0.1003	0.0904	0.4240	7.41	126.42	101.325	0.1003	0.0924	0.3833	6.10
90.39	40.00	0.2005	0.1868	0.6075	6.74	117.84	101.325	0.2005	0.1889	0.5726	5.75
86.14	40.00	0.3008	0.2870	0.7041	5.91	112.64	101.325	0.3008	0.2885	0.6795	5.23
83.20	40.00	0.3999	0.3879	0.7640	5.11	108.89	101.325	0.3999	0.3888	0.7479	4.66
81.08	40.00	0.5000	0.4906	0.8075	4.35	106.27	101.325	0.5000	0.4911	0.7975	4.08
79.24	40.00	0.6007	0.5940	0.8426	3.66	104.00	101.325	0.6007	0.5942	0.8368	3.50
78.03	40.00	0.7131	0.7093	0.8774	2.93	102.48	101.325	0.7131	0.7094	0.8746	2.86
77.10	40.00	0.8103	0.8084	0.9086	2.36	101.31	101.325	0.8103	0.8084	0.9071	2.31
76.54	40.00	0.8998	0.8991	0.9444	1.91	100.64	101.325	0.8998	0.8991	0.9431	1.86
76.27	40.00	1.0000	1.0000	0.9990	1.00	100.36	101.325	1.0000	1.0000	1.0000	1.00

Table 2. Antoine Equation Constants and Derived *Pxy* Data at 75 °C for Water (1) + Acrylic Acid (2)^a

<i>x</i> ₁	<i>A</i> (<i>x</i> ₁)	<i>B</i> (<i>x</i> ₁)	<i>C</i> (<i>x</i> ₁)	<i>P</i> (kPa)	<i>y</i> ₁ (calc)	α (calc)
0.0000	6.351301	1443.082	189.614	7.90	0.0000	
0.0888	6.068280	1293.193	191.902	16.71	0.4701	9.11
0.1855	6.331621	1389.949	203.472	21.89	0.6318	7.53
0.2862	6.372183	1366.787	200.376	25.64	0.7152	6.26
0.3875	6.470337	1382.736	200.819	28.65	0.7685	5.25
0.4905	6.551704	1403.820	202.531	31.15	0.8088	4.39
0.5940	6.692770	1463.838	208.311	33.56	0.8425	3.66
0.7094	6.672208	1434.742	204.974	35.29	0.8772	2.93
0.8084	6.794623	1491.634	210.166	36.63	0.9086	2.36
0.8991	6.767238	1468.217	207.708	37.48	0.9446	1.91
1.0000	6.900703	1547.463	215.771	37.91	1.0000	1.00

^a $\log P(x_1) = A(x_1) - B(x_1)/[t + C(x_1)]$; $\log$ = base 10, $P(x_1)$ = kPa, t = °C.

equation that describes thermodynamic equilibrium between the phases at *T* and *P* is

$$\phi_i y_i P = x_i \gamma_i P_i^0 \exp[(P - P_i^0) V_i / (RT)] \quad i = 1, 2 \quad (1)$$

where ϕ is the vapor-phase fugacity coefficient, γ is the liquid-phase activity coefficient (reference state = pure liquid at system *T* and *P*), P_i^0 is the pure-component vapor pressure, and V_i is the pure-component saturated-liquid molar volume. Saturated-liquid volume data were taken from the DIPPR databank,¹⁵ and

the vapor-phase fugacity coefficients were computed from the second-virial equation of state. Second-virial coefficients were estimated from the Hayden–O'Connell correlation.¹⁷

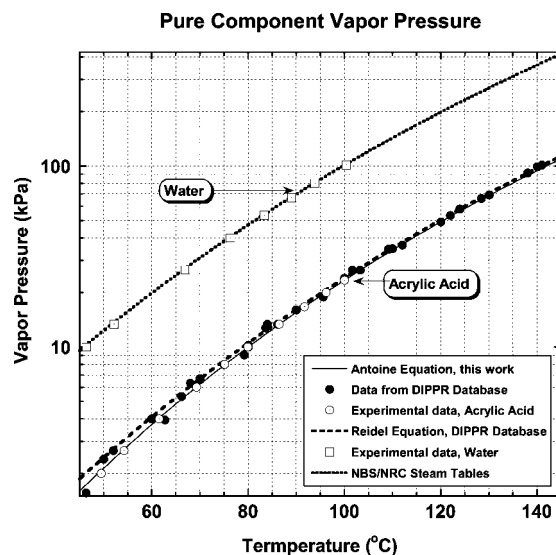


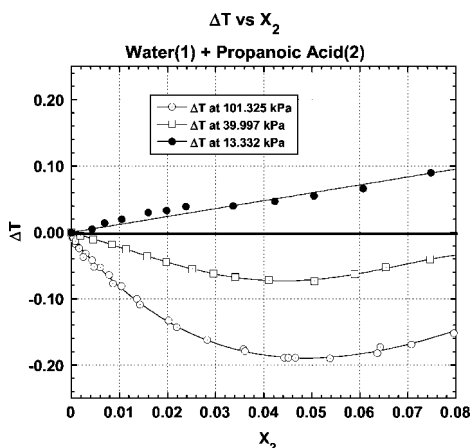
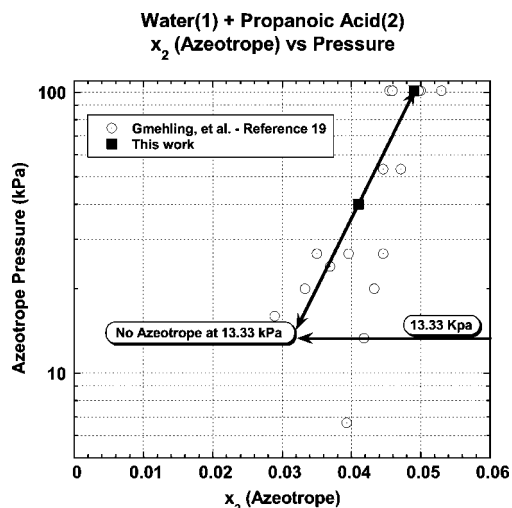
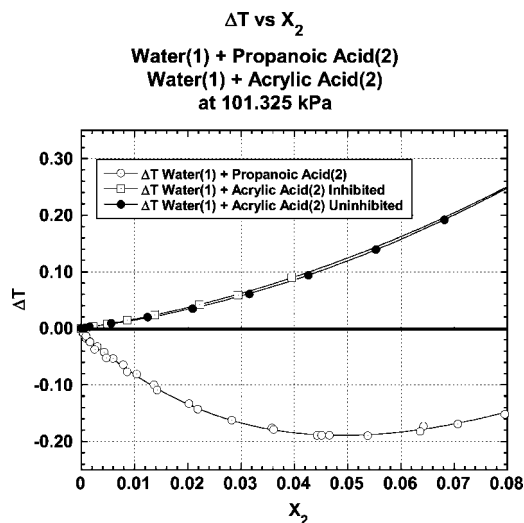
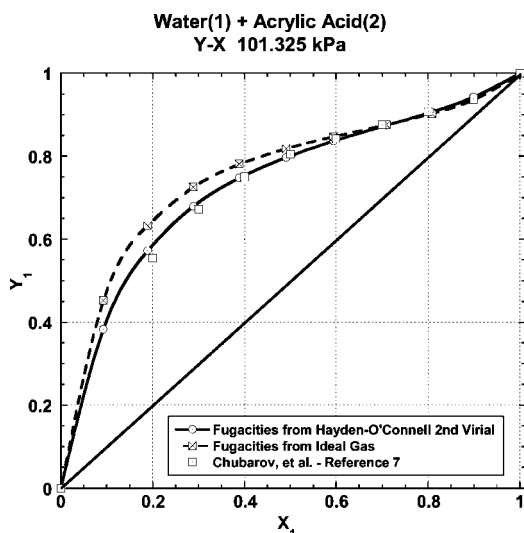
Figure 1. Comparison of measured pure-component vapor-pressure data with literature data.

Table 3. UNIQUAC Parameters from Barker's Method Fit for Water (1) + Acrylic Acid (2)

data set	UNIQUAC τ_{12}	UNIQUAC τ_{21}	RMS error % P
all measured isobaric data	0.50571	0.51123	1.59
derived isothermal data at 50, 75, 100, and 120 °C	0.12136	1.38309	1.61
derived isothermal data at 50 °C	0.11525	1.40114	1.79
derived isothermal data at 75 °C	0.13050	1.35842	1.75
derived isothermal data at 100 °C	0.13023	1.36357	1.55
derived isothermal data at 120 °C	0.11055	1.40576	1.30

The fitted G^E equation is the UNIQUAC model; details of the equation and the derived activity coefficient equations are described elsewhere.¹⁸ Table 3 presents results from fitting the UNIQUAC model to the following: (1) all of the experimental PTx data (including the dilute solution data discussed below) and (2) the derived Px data at 50, 75, 100, and 120 °C. The UNIQUAC fits to the derived isothermal Px data were analyzed to produce G^E data that could be numerically differentiated and used with the Gibbs–Helmholtz equation to predict H^E and TS^E as described in the Discussion section.

Table 4 contains dilute-solution data for water + propanoic acid and water + acrylic acid. These data indicate the presence or absence of an azeotrope from examination for a Gibbs–Konovalov extremum: a maximum or minimum in the isobaric

**Figure 2.** Dilute-solution data for water (1) + propanoic acid (2); $\Delta T = [T - T(x_2 = 0)]$.**Figure 3.** Azeotropic composition vs pressure for water (1) + propanoic acid (2).**Figure 4.** Dilute-solution data for water (1) + propanoic acid (2) and water (1) + acrylic acid (2) at 101.325 kPa; $\Delta T = [T - T(x_2 = 0)]$.**Figure 5.** Importance of the Hayden–O’Connell second virial coefficient correlation to calculate fugacity coefficients for associating acid mixtures.

Tx bubble-point curve. Figure 2 shows ΔT vs acid mole fraction, x_2 , data for water + propanoic acid. Analysis of these data give azeotropes at $[P = 101.325 \text{ kPa}, x_2 = 0.049]$ and $[P = 39.997 \text{ kPa}, x_2 = 0.041]$ and a zeotrope at $[P = 13.332 \text{ kPa}]$. These data are compared to literature data¹⁹ for azeotropes of water + propanoic acid in Figure 3.

Figure 4 shows ΔT vs x_2 data for water + acrylic acid at 101.325 kPa compared to the data for water + propanoic acid. These data show conclusively that no Tx extremum, and hence no azeotrope, exists for water + acrylic acid at 101.325 kPa.

Discussion

Use of the Hayden–O’Connell second-virial coefficient correlation to calculate vapor-phase fugacity coefficients and compositions is essential for systems that contain associating organic acids. Figure 5 shows the derived 101.325 kPa water + acrylic acid x – y phase diagrams for Hayden–O’Connell vapor compositions vs vapor compositions from an ideal-gas vapor-phase model. Also shown are measured xy data from Chubarov et al.⁷ Note the large deviations in y in the acrylic acid-rich end of the diagram compared to the water-rich end of the diagram.

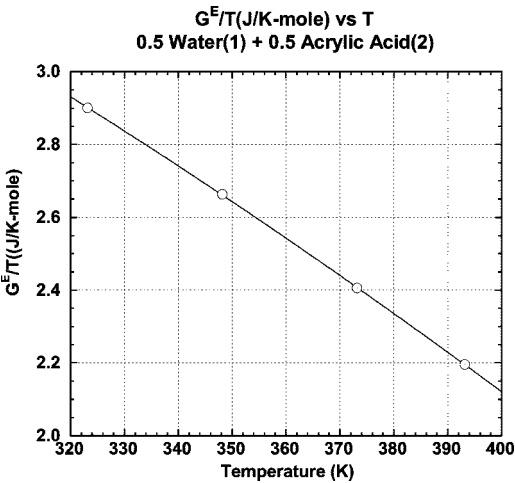


Figure 6. Equimolar G^E/T vs temperature for water (1) + acrylic acid (2).

Excess mixture thermodynamic data derived from VLE measurements can provide insight into the molecular nature of the system.^{20–22} The Gibbs–Helmholtz equation was used to derive enthalpy-of-mixing, H^E , data from the temperature dependence of the G^E data,

$$H^E = -T^2(\partial g/\partial T)_x \tag{2}$$

Table 4. Dilute-Solution PTx Data for Water (1) + Propanoic Acid (2) and Water (1) + Acrylic Acid (2)

P (kPa)	z_2	x_2	T (°C)	ΔT (°C)	P (kPa)	z_2	x_2	T (°C)	ΔT (°C)
Water (1) + Propanoic Acid (2)									
101.325	0.00000	0.00000	99.979	0.000	101.325	0.07928	0.07952	99.827	−0.152
101.325	0.00007	0.00007	99.978	−0.001	101.325	0.08951	0.08984	99.862	−0.117
101.325	0.00026	0.00026	99.971	−0.008	101.325	0.10821	0.10873	99.921	−0.058
101.325	0.00041	0.00040	99.971	−0.008	40.00	0.00000	0.00000	75.943	0.000
101.325	0.00054	0.00053	99.967	−0.012	40.00	0.00188	0.00187	75.938	−0.005
101.325	0.00087	0.00087	99.966	−0.013	40.00	0.00454	0.00452	75.932	−0.011
101.325	0.00091	0.00090	99.961	−0.018	40.00	0.00843	0.00840	75.925	−0.018
101.325	0.00152	0.00151	99.956	−0.023	40.00	0.01141	0.01138	75.918	−0.025
101.325	0.00166	0.00165	99.955	−0.024	40.00	0.01584	0.01580	75.907	−0.036
101.325	0.00250	0.00248	99.942	−0.037	40.00	0.01975	0.01972	75.898	−0.045
101.325	0.00299	0.00296	99.947	−0.032	40.00	0.02521	0.02518	75.888	−0.055
101.325	0.00432	0.00428	99.937	−0.042	40.00	0.02952	0.02949	75.881	−0.062
101.325	0.00476	0.00472	99.927	−0.052	40.00	0.03414	0.03412	75.876	−0.067
101.325	0.00605	0.00600	99.926	−0.053	40.00	0.04108	0.04109	75.871	−0.072
101.325	0.00790	0.00784	99.915	−0.064	40.00	0.05054	0.05058	75.870	−0.073
101.325	0.00868	0.00862	99.902	−0.077	40.00	0.05882	0.05890	75.880	−0.063
101.325	0.01049	0.01042	99.898	−0.081	40.00	0.06528	0.06539	75.890	−0.053
101.325	0.01370	0.01362	99.879	−0.100	40.00	0.07446	0.07463	75.902	−0.041
101.325	0.01436	0.01428	99.870	−0.109	13.33	0.00000	0.00000	51.867	0.000
101.325	0.02027	0.02019	99.846	−0.133	13.33	0.00429	0.00429	51.872	0.005
101.325	0.02195	0.02186	99.836	−0.143	13.33	0.00692	0.00692	51.881	0.014
101.325	0.02835	0.02827	99.817	−0.162	13.33	0.01050	0.01050	51.887	0.020
101.325	0.03586	0.03579	99.803	−0.176	13.33	0.01594	0.01595	51.897	0.030
101.325	0.03619	0.03612	99.800	−0.179	13.33	0.01976	0.01978	51.900	0.033
101.325	0.04428	0.04425	99.790	−0.189	13.33	0.02379	0.02382	51.906	0.039
101.325	0.04512	0.04510	99.790	−0.189	13.33	0.03355	0.03362	51.907	0.040
101.325	0.04662	0.04660	99.790	−0.189	13.33	0.04217	0.04227	51.914	0.047
101.325	0.05371	0.05374	99.789	−0.190	13.33	0.05029	0.05044	51.922	0.055
101.325	0.06350	0.06361	99.797	−0.182	13.33	0.06044	0.06065	51.933	0.066
101.325	0.06411	0.06422	99.806	−0.173	13.33	0.07442	0.07473	51.957	0.090
101.325	0.07054	0.07070	99.810	−0.169					
Water (1) + Acrylic Acid (2) [Inhibited with 0.1% Phenothiazine]									
101.325	0.00000	0.00000	99.980	0.000	101.325	0.01379	0.01382	100.004	0.024
101.325	0.00227	0.00227	99.983	0.003	101.325	0.02210	0.02215	100.022	0.042
101.325	0.00495	0.00495	99.988	0.008	101.325	0.02933	0.02940	100.039	0.059
101.325	0.00859	0.00860	99.995	0.015	101.325	0.03940	0.03952	100.070	0.090
Water (1) + Acrylic Acid (2) [Uninhibited]									
101.325	0.00000	0.00000	99.980	0.000	101.325	0.02086	0.02090	100.015	0.035
101.325	0.00074	0.00074	99.981	0.001	101.325	0.03146	0.03154	100.041	0.061
101.325	0.00150	0.00150	99.983	0.003	101.325	0.04249	0.04263	100.074	0.094
101.325	0.00560	0.00561	99.989	0.009	101.325	0.05507	0.05527	100.119	0.139
101.325	0.01241	0.01243	100.000	0.020	101.325	0.06785	0.06814	100.172	0.192

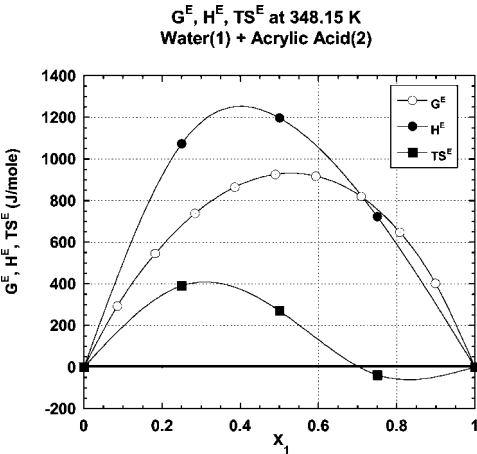


Figure 7. Thermodynamic excess functions, G^E , H^E , and TS^E , vs composition for water (1) + acrylic acid (2) at 348.15 K.

where $g = G^E/T$. Figure 6 shows equimolar g vs T from analysis of our PTx data. The slope of the curve at 75 °C gives an equimolar H^E of 1197 J/mol. Because H^E is derived by differentiation, experimental error is magnified. The equimolar excess thermodynamic functions at 75 °C derived from the PTx measurements are as follows: $G^E = 927$ J/mol, $H^E = 1197$ J/mol, $TS^E = 270$ J/mol, and $S^E = 0.78$ J/K·mol ($TS^E = H^E -$

G^E). Figure 7 shows the derived thermodynamic excess functions vs composition.

Malesiński²¹ classified binary mixtures according to the signs of their excess functions. He relates this classification to the like interactions (1–1, 2–2) and the unlike interactions (1–2) of the two molecules in solution. As shown in Figure 7, the system water + acrylic acid gives $G^E > 0$, $H^E > 0$, and $S^E > 0$ except when $x_1 > 0.7$. Malesiński's classification system suggests that negative values of TS^E in the water-rich end of the water + acrylic acid system may not be due to strong interactions of the (1–2) type but rather due to persistence of strong orientational effects between the associating acid molecules in the solution. However, O'Connell et al.,²² in a study of over 300 systems, found that it was equally common for S^E to be positive or negative when G^E and H^E are positive for associating systems. Finally, in an analogous water + organic acid system (albeit a saturated organic acid), the system water + acetic acid at 25 °C²³ shows $G^E > H^E > 0$ (except for a small $H^E < 0$ region near the pure water end) and $TS^E < 0$.

There are no water + acrylic acid H^E data in the literature that could be used for Gibbs–Helmholtz thermodynamic consistency²⁴ testing of the PTx data presented here. Two sets of H^E data for water + propanoic acid are given in the DECHEMA Heats of Mixing Data Collection²⁵ at temperatures of 8²⁶ and 25 °C;²⁷ the H^E data²⁷ at 25 °C show a large positive H^E , which is skewed similar to the derived H^E data for acrylic acid in Figure 7.

Additional experiments to further explain the mixture thermodynamics of water + acrylic acid would include the following: measured H^E vs composition at several temperatures, C_p^E measured by differential scanning calorimetry,⁴ and V^E measured by vibrating-tube densimetry.^{1,3} Molecular simulation studies could also test the signs of the excess functions reported here.

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